CEBU INSTITUTE OF TECHNOLOGY – UNIVERSITY

College of Computer Studies • Department of Information Technology
IT332 – Capstone Project 2: Systems Development & Research Validation

REQUIREMENTS INVENTORY & REQUIREMENTS TRACEABILITY MATRIX (RTM)

Formal Specification Derived from Updated SRS v2.0, Updated SDD v3.1, and Software Test Document (STD v2.0)

System: Task Force Bruno (Campus Pet Management Portal)
Team: Team 18 (Task Force Bruno)
Document Version: 2.0 (Post-Refactoring Release)
Institutional Scope: CIT-U Campus Premises & RA 9482 Compliance
Total Requirements: 21 Requirements Active
Verification Status: 100% Passed (STD v2.0 Verified)

1. OVERVIEW & TRACEABILITY METHODOLOGY

This document serves as the formal engineering catalog and verification bridge for **Task Force Bruno: Integrated Campus Pet Management System** at **Cebu Institute of Technology – University (CIT-U)**. Prepared in strict accordance with Capstone 2 standards, this submission couples two fundamental systems engineering artifacts:

1. Requirements Inventory (Catalog): Defines each institutional requirement derived from the project's SMART goals, detailing the exact operational transaction (*Input* ➔ *Process* ➔ *Output*), categorization, priority, status percentage, and subsystem ownership.
2. Requirements Traceability Matrix (RTM): Establishes bidirectional traceability connecting each requirement through the engineering lifecycle: from *Updated SRS v2.0* specifications (Use Cases, Activity Diagrams, Wireframes) to *Updated SDD v3.1* architecture (Class Diagrams, Sequence Diagrams, User Interfaces, Database Schemas), software implementation components, and final *STD v2.0* test verification cases.

2. REQUIREMENTS INVENTORY (FUNCTIONAL & NON-FUNCTIONAL SPECIFICATIONS)

Table 1 itemizes the 21 active system requirements categorized by primary SMART Goal ID (G1–G6) and transaction lifecycle:

Goal ID	SMART Goal	Req. ID	Requirement Statement	Transaction (Input ➔ Process ➔ Output)	Type	Priority	Status %	Notes / Issues	Owner
G1	**Enable rapid lookup of companion identity, vaccina...**	**REQ-101**	The system shall provide unauthenticated public access to the Emergency Pet Passport upon scanning a physical QR collar tag.	User points camera at QR tag → System captures matrix and resolves pet ID → Browser displays emergency contacts and rabies vaccination badge without requiring authentication.	Functional	Must	100%	Verified with Bruno PET-001; public passport renders in < 1.0s.	Mobile / Frontend Lead
G1	**Enable rapid lookup of companion identity, vaccina...**	**REQ-102**	The system shall provide a client-side static image QR decoding fallback when hardware camera permissions are denied or unavailable.	Camera permission denied / user selects file → System accepts static image upload and decodes matrix via canvas jsQR → Pet profile resolved without server roundtrip.	Functional	Must	100%	Resolved post-MVP (INC-02); tested on Android WebKit and iOS Safari.	Frontend Lead
G1	**Enable rapid lookup of companion identity, vaccina...**	**REQ-103**	The system shall allow users to search campus pet profiles using natural language physical trait descriptors and location heuristics.	User inputs search query (e.g. 'calico white paws canteen') → PetAISearchAPIView tokenizes descriptors → Grid of matching companion profiles returned.	Functional	Should	100%	Heuristic search active; filters by coat pattern, color, and landmark.	Backend Lead

Goal ID	SMART Goal	Req. ID	Requirement Statement	Transaction (Input ➔ Process ➔ Output)	Type	Priority	Status %	Notes / Issues	Owner
G2	**Digitize 100% of campus animal sightings with phot...**	**REQ-201**	The system shall capture crowdsourced stray animal sightings with mandatory photo upload and standardized CIT-U campus landmark selection.	Student fills sighting form and selects campus landmark from dropdown → System uploads image to Supabase and saves record → New feed item published.	Functional	Must	100%	Replaced free text with standardized 8-zone landmark dictionary (INC-05).	Frontend / Backend Lead
G2	**Digitize 100% of campus animal sightings with phot...**	**REQ-202**	The system shall provide an Emergency Sentinel priority selector for sightings of injured animals, sick strays, or active bite risks.	User toggles 'Emergency: Bite / Scratch Risk' → System displays immediate RA 9482 first-aid guidance and flags post with urgent red banner and clinic hotline → High-priority alert broadcast on campus feed.	Functional	Must	100%	Provides immediate 15-minute soap/water wash instructions and direct MDC hotline link.	Full-Stack Lead
G2	**Digitize 100% of campus animal sightings with phot...**	**REQ-203**	The system shall enable authorized staff to admit verified stray sightings into official pet registration via a 1-click state handoff.	Staff clicks 'Admit & Intake Stray' on sighting card → System routes to Add New Pet form auto-populating photo, species, and landmark → Staff verifies and saves pet record without manual re-typing.	Functional	Must	100%	Eliminates 100% of redundant data entry, saving 4-6 minutes per intake.	Frontend Lead
G3	**Enforce compliance with Republic Act No. 9482 thro...**	**REQ-301**	The system shall allow authorized MDC clinic staff to log clinical treatment records, routine veterinary visits, and anti-rabies vaccination dates.	Clinic staff submits vaccination record and next due date → System updates pet medical log and refreshes vaccination validity badge → Pet profile displays 'Fully Vaccinated'.	Functional	Must	100%	Restricted to authenticated MDC personnel; updates live badge chips.	Clinic Lead / Backend
G3	**Enforce compliance with Republic Act No. 9482 thro...**	**REQ-302**	The system shall automatically calculate and display an active 14-day observation quarantine countdown badge upon logging an animal bite or scratch incident.	Clinic staff submits bite incident record → System computes statutory expiry date (incident date + 14 days) and sets quarantine flag → Profile displays red quarantine alert banner with live remaining days/hours counter.	Functional	Must	100%	Statutory compliance with Republic Act No. 9482 Section 5 biosecurity protocols.	Full-Stack Lead
G3	**Enforce compliance with Republic Act No. 9482 thro...**	**REQ-303**	The system shall restrict all clinical write endpoints strictly to authenticated users with the MDC_STAFF or ADMIN role.	Incoming API request to /api/medical/* → DRF RBAC permission class verifies caller JWT role claim → Grants write access if MDC staff; otherwise returns HTTP 403 Forbidden.	Functional	Must	100%	Gated by JWT claims; unauthorized student requests blocked at gateway.	Security / Backend Lead
G4	**Provide clear inventory accountability broken down...**	**REQ-401**	The system shall provide a multi-step digital adoption application form capturing applicant background, residence type, and pet care experience.	User completes adoption form and submits application → System validates required fields and sets status to 'Pending' → Application queued in Administrative Review Dashboard.	Functional	Must	100%	Includes validation on contact numbers, landlord permissions, and residence.	Frontend Lead
G4	**Provide clear inventory accountability broken down...**	**REQ-402**	The system shall allow authorized welfare staff to review, approve, or reject pending adoption applications with automated status mutation on the pet profile.	Staff reviews application in Shelter Dashboard and clicks 'Approve' → System changes application state to 'Approved' and pet status to 'Adopted' → Public adoption gallery updates in real-time.	Functional	Must	100%	Automates pet status transition upon administrative approval.	Backend / Staff Lead

Goal ID	SMART Goal	Req. ID	Requirement Statement	Transaction (Input ➔ Process ➔ Output)	Type	Priority	Status %	Notes / Issues	Owner
G4	**Provide clear inventory accountability broken down...**	**REQ-403**	The system shall track supply inventory intake and disbursements strictly categorized by standardized operational physical units (kg, cans, vials, pieces).	Staff logs supply donation with quantity and enum unit → System updates active stock ledger and creates an immutable audit trail transaction record → Real-time stock levels displayed on dashboard.	Functional	Must	100%	Strict Django TextChoices enum prevents arbitrary unit entries.	Backend / Inventory Lead
G4	**Provide clear inventory accountability broken down...**	**REQ-404**	The system shall enforce boundary checks preventing disbursements from exceeding active on-hand quantities, preventing negative inventory.	Staff attempts to deduct quantity exceeding available stock → System evaluates quantity constraint and blocks deduction → UI displays insufficient stock warning modal.	Functional	Must	100%	Boundary validation verified in TC-SUP-003; zero negative stock entries.	Backend Lead
G5	**Achieve high learnability, low cognitive load, vis...**	**REQ-501**	The system shall provide an aggregated community newsfeed displaying pet welfare posts, sighting telemetry, likes, comments, and campus companion counts.	User accesses homepage → System fetches newsfeed stream and live telemetry aggregates → Render interactive timeline with like and comment interactions.	Functional	Must	100%	Includes live telemetry metrics bar and community reaction toggles.	Frontend Lead
G5	**Achieve high learnability, low cognitive load, vis...**	**REQ-502**	The system shall provide an interactive 4-step onboarding walkthrough (UserGuideModal.jsx) triggered on first login and accessible via the persistent navigation bar.	First-time user opens portal → System checks localStorage flag and launches walkthrough modal explaining QR scanning, search, sightings, and adoption → Dismissal sets completion flag.	Functional	Must	100%	Directly targets SUS score uplift (raising baseline from 62.32 to ≥ 80.0).	UI / UX Lead
G5	**Achieve high learnability, low cognitive load, vis...**	**REQ-503**	The system shall enforce institutional account registration restricted exclusively to @cit.edu email addresses and XX-XXXX-XXX ID number format.	User enters email and ID → System validates institutional domain and regex syntax → Creates account with hashed password and returns authenticated JWT session payload.	Functional	Must	100%	Enforces automatic JWT session handoff post-registration (resolved INC-03).	Security / Auth Lead
G5	**Achieve high learnability, low cognitive load, vis...**	**REQ-504**	The system shall maintain dynamic mobile viewport layouts (100dvh) with input auto-scrolling to prevent virtual keyboard occlusion of form elements.	Mobile user taps into password input field → Layout dynamically resizes and triggers smooth scroll to center → Password field, eye visibility toggle, and submit button remain fully visible.	Non-functional	Must	100%	Resolved mobile keyboard overlap on 375px screens (INC-01).	UI / UX Lead
G5	**Achieve high learnability, low cognitive load, vis...**	**REQ-505**	The system shall automatically terminate user sessions and purge stored tokens after 15 minutes of continuous user inactivity on client terminals.	Inactivity timer detects 15 minutes of zero user interaction → Client removes JWT token from localStorage and revokes session → Redirects to login with timeout notice.	Non-functional	Must	100%	Protects public campus workstation terminals from session hijacking.	Security Lead

Goal ID	SMART Goal	Req. ID	Requirement Statement	Transaction (Input ➔ Process ➔ Output)	Type	Priority	Status %	Notes / Issues	Owner
G6	**Ensure sub-500ms client asset generation, 99.5% op...**	**REQ-601**	The system shall render a client-side printable 300 DPI QR collar tag graphic badge in under 500 milliseconds.	User clicks 'Generate Collar Tag QR' on pet profile → HTML5 canvas renders 1200x1200px graphic with CIT-U Maroon border and QR matrix → File downloaded as PNG in < 500ms.	Non-functional	Must	100%	Canvas generation completed in 185ms (315ms buffer under threshold).	Frontend Lead
G6	**Ensure sub-500ms client asset generation, 99.5% op...**	**REQ-602**	The system shall comply with WCAG 2.1 AA color contrast standards across all telemetry badges and status chips (Green for vaccinated, Red for quarantine, Amber for pending).	System renders status badges on profiles and tables → Visual contrast ratio meets or exceeds 4.5:1 against card backgrounds → Clear legibility across mobile displays.	Non-functional	Must	100%	Contrast verified across mobile displays and dark/light backgrounds.	UI / UX Lead
G6	**Ensure sub-500ms client asset generation, 99.5% op...**	**REQ-603**	The system shall maintain 99.5% operational uptime during university operating hours with automated daily Supabase database backups.	Production instance operates under monitoring watchdog → Supabase automated schedule runs nightly snapshot → Service continuously accessible via HTTPS/TLS 1.3.	Non-functional	Must	100%	Hosted on Supabase cloud PostgreSQL with SSL/TLS encryption.	DevOps / Backend Lead

3. REQUIREMENTS TRACEABILITY MATRIX (RTM)

Table 2 maps every active requirement to its formal design artifacts in the Updated SRS v2.0 and SDD v3.1, its physical implementation modules, and its execution outcome in the Software Test Document (STD v2.0):

Requirement ID	SRS Reference (Use Cases, Activity Diagrams, Wireframes)	SDD Reference (Class Diagrams, Sequence Diagrams, UIs, DBs)	Implementation Module(s) / Component(s)	STD Reference (Test Case IDs)	Status
REQ-101	UC-1.1, AD-1.1, WF-1.1	CD-1.1, SD-1.1, UI-1.1, DB-1.1	`CollarQRScanner.jsx, views.py`	**TC-QR-001**	Passed
REQ-102	UC-1.1, AD-1.1, WF-1.1	CD-1.1, SD-1.1, UI-1.1, DB-1.1	`CollarQRScanner.jsx`	**TC-QR-002**	Passed
REQ-103	UC-1.3, AD-1.3, WF-1.3	CD-1.3, SD-1.3, UI-1.3, DB-1.3	`PetAISearchAPIView, views.py`	**TC-REG-003**	Passed
REQ-201	UC-3.2, AD-3.2, WF-3.2	CD-3.2, SD-3.2, UI-3.2, DB-3.2	`ReportSightingView.jsx, views.py`	**TC-SGT-001**	Passed
REQ-202	UC-3.2, AD-3.2, WF-3.2	CD-3.2, SD-3.2, UI-3.2, DB-3.2	`ReportSightingView.jsx, EmergencyProtocolModal.jsx`	**TC-SGT-002**	Passed
REQ-203	UC-1.3, AD-1.3, WF-1.3	CD-1.3, SD-1.3, UI-1.3, DB-1.3	`AddNewPet.jsx, ReportSightingView.jsx`	**TC-SGT-003, TC-REG-001**	Passed
REQ-301	UC-1.2, AD-1.2, WF-1.2	CD-1.2, SD-1.2, UI-1.2, DB-1.2	`MedicalLogs.jsx, views.py`	**TC-MED-001**	Passed
REQ-302	UC-1.2, AD-1.2, WF-1.2	CD-1.2, SD-1.2, UI-1.2, DB-1.2	`MedicalLogs.jsx, views.py`	**TC-MED-002**	Passed
REQ-303	UC-1.2, AD-1.2, WF-1.2	CD-1.2, SD-1.2, UI-1.2, DB-1.2	`MDCAuthGate.jsx, views.py`	**TC-MED-003, TC-SEC-001**	Passed
REQ-401	UC-2.1, AD-2.1, WF-2.1	CD-2.1, SD-2.1, UI-2.1, DB-2.1	`AdoptionGallery.jsx, views.py`	**TC-ADPT-001, TC-ADPT-002**	Passed
REQ-402	UC-2.1, AD-2.1, WF-2.1	CD-2.1, SD-2.1, UI-2.1, DB-2.1	`StaffDashboard.jsx, views.py`	**TC-ADPT-003**	Passed
REQ-403	UC-2.2, AD-2.2, WF-2.2	CD-2.2, SD-2.2, UI-2.2, DB-2.2	`InventoryView.jsx, models.py, views.py`	**TC-SUP-001, TC-SUP-002**	Passed
REQ-404	UC-2.2, AD-2.2, WF-2.2	CD-2.2, SD-2.2, UI-2.2, DB-2.2	`InventoryView.jsx, views.py`	**TC-SUP-003**	Passed
REQ-501	UC-3.1, AD-3.1, WF-3.1	CD-3.1, SD-3.1, UI-3.1, DB-3.1	`NewsfeedView.jsx, views.py`	**TC-FEED-001**	Passed
REQ-502	UC-4.1, AD-4.1, WF-4.1	CD-4.1, SD-4.1, UI-4.1, DB-4.1	`UserGuideModal.jsx, Navbar.jsx`	**TC-ONB-001**	Passed
REQ-503	UC-4.2, AD-4.2, WF-4.2	CD-4.2, SD-4.2, UI-4.2, DB-4.2	`RegisterUserView.jsx, views.py`	**TC-AUTH-001, TC-AUTH-002**	Passed
REQ-504	UC-4.2, AD-4.2, WF-4.2	CD-4.2, SD-4.2, UI-4.2	`LoginForm.jsx`	**TC-AUTH-003**	Passed
REQ-505	UC-4.2, AD-4.2, WF-4.2	CD-4.2, SD-4.2, DB-4.2	`JWTAuthMiddleware, App.jsx`	**TC-AUTH-004**	Passed
REQ-601	UC-1.3, AD-1.3, WF-1.3	CD-1.3, SD-1.3, UI-1.3	`AddNewPet.jsx, canvasTagGenerator.js`	**TC-REG-002, TC-PERF-001**	Passed
REQ-602	Section 3.4 (Usability)	Section 1.3 (Style Guide)	`Global CSS, StatusChip.jsx`	**TC-USA-001**	Passed
REQ-603	Section 3.4 (Reliability)	Architectural Design	`render.yaml, Supabase Cloud`	**TC-SEC-001**	Passed

4. TRACEABILITY AUDIT & ENGINEERING SIGN-OFF

Every requirement listed has been traced to a concrete implementation component and verified via comprehensive test execution documented in the Software Test Document (STD v2.0). All defect incidents identified during MVP validation (INC-01 through INC-05) have been systematically resolved and certified:

Role	Name & Designation	Signature	Date of Audit
Lead Systems Architect	Task Force Bruno Development Lead	_____	October 10, 2026
QA & Verification Lead	Team 18 QA Specialist	_____	October 10, 2026
Clinical Stakeholder	CIT-U Medical-Dental Clinic (MDC) Rep	_____	October 10, 2026
Capstone Project Adviser	College of Computer Studies Faculty	_____	October 10, 2026

— End of Traceability Matrix & Requirements Inventory —

Task Force Bruno Capstone Engineering & Research Team
Department of Information Technology • Cebu Institute of Technology – University